dataset (Elsahar et al., 2018) and is much larger than Google-RE with a broader set of relations. We consider 41 Wikidata relations and subsample at most 1000 facts per relation. As with the Google-RE corpus, we manually define a template for each relation (see Table 3 for some examples). In contrast to the Google-RE knowledge source, T-REx facts were automatically aligned to Wikipedia and hence this alignment can be noisy. However, Elsahar et al. (2018) report an accuracy of 97.8% for the alignment technique over a test set.

4.1.3 ConceptNet

ConceptNet (Speer and Havasi, 2012) is a multilingual knowledge base, initially built on top of Open Mind Common Sense (OMCS) sentences. OMCS represents commonsense relationships between words and/or phrases. We consider facts from the English part of ConceptNet that have single-token objects covering 16 relations. For these ConceptNet triples, we find the OMCS sentence that contains both the subject and the object. We then mask the object within the sentence and use the sentence as template for querying language models. If there are several sentences for a triple, we pick one at random. Note that for this knowledge source there is no explicit alignment of facts to Wikipedia sentences.

4.1.4 SQuAD

SQuAD (Rajpurkar et al., 2016) is a popular question answering dataset. We select a subset of 305 context-insensitive questions from the SQuAD development set with single token answers. We manually create cloze-style questions from these questions, *e.g.*, rewriting "Who developed the theory of relativity?" as "The theory of relativity was developed by ____". For each question and answer pair, we know that the corresponding fact is expressed in Wikipedia since this is how SQuAD was created.

4.2 Models

We consider the following pretrained case-sensitive language models in our study (see Table 1): fairseq-fconv (*Fs*), Transformer-XL large (*Txl*), ELMo original (*Eb*), ELMo 5.5B (*E5B*), BERT-base (*Bb*) and BERT-large (*Bl*). We use the natural way of generating tokens for each model by following the definition of the training objective function.

Assume we want to compute the generation for the token at position t. For unidirectional language models, we use the network output ($\mathbf{h}_{t-1}$) just before the token to produce the output layer softmax. For ELMo we consider the output just before ($\overrightarrow{\mathbf{h}}_{t-1}$) for the forward direction and just after ($\overleftarrow{\mathbf{h}}_{t+1}$) for the backward direction. Following the loss definition in (Peters et al., 2018a), we average forward and backward probabilities from the corresponding softmax layers. For BERT, we mask the token at position t, and we feed the output vector corresponding to the masked token ($\mathbf{h}_t$) into the softmax layer. To allow a fair comparison, we let models generate over a unified vocabulary, which is the intersection of the vocabularies for all considered models (~21K case-sensitive tokens).

4.3 Baselines

To compare language models to canonical ways of using off-the-shelf systems for extracting symbolic knowledge and answering questions, we consider the following baselines.

Freq: For a subject and relation pair, this baseline ranks words based on how frequently they appear as objects for the given relation in the test data. It indicates the upper bound performance of a model that always predicts the same objects for a particular relation.

RE: For the relation-based knowledge sources, we consider the pretrained Relation Extraction (RE) model of Sorokin and Gurevych (2017). This model was trained on a subcorpus of Wikipedia annotated with Wikidata relations. It extracts relation triples from a given sentence using an LSTM-based encoder and an attention mechanism. Based on the alignment information from the knowledge sources, we provide the relation extractor with the sentences known to express the test facts. Using these datasets, RE constructs a knowledge graph of triples. At test time, we query this graph by finding the subject entity and then rank all objects in the correct relation based on the confidence scores returned by RE. We consider two versions of this procedure that differ in how the entity linking is implemented: $\mathbf{RE}_n$ makes use of a naïve entity linking solution based on exact string matching, while $\mathbf{RE}_o$ uses an oracle for entity linking in addition to string matching. In other words, assume we query for the object o of a test subject-relation fact (s, r, o) expressed in a sentence x. If RE has extracted any triple (s', r, o') from that sentence x, s' will be linked to s and o' to o. In practice, this means RE can return the correct solution o if *any* relation instance of the right type was extracted from x, regardless of whether it has a wrong subject or object.

DrQA: Chen et al. (2017) introduce DrQA, a popular system for open-domain question answering. DrQA predicts answers to natural language questions using a two step pipeline. First, a TF/IDF information retrieval step is used to find relevant articles from a large store of documents (e.g. Wikipedia). On the retrieved top k articles, a neural reading comprehension model then extracts answers. To avoid giving the language models a competitive advantage, we constrain the predictions of DrQA to single-token answers.

4.4 Metrics

We consider rank-based metrics and compute results per relation along with mean values across all relations. To account for multiple valid objects for a subject-relation pair (*i.e.*, for N-M relations), we follow Bordes et al. (2013) and remove from the candidates when ranking at test time all other valid objects in the training data other than the one we test. We use the mean precision at k ($P@k$). For a given fact, this value is 1 if the object is ranked among the top k results, and 0 otherwise.

4.5 Considerations

There are several important design decisions we made when creating the LAMA probe. Below we give more detailed justifications for these decisions.

Manually Defined Templates For each relation we manually define a template that queries for the object slot in that relation. One can expect that the choice of templates has an impact on the results, and this is indeed the case: for some relations we find both worse and better ways to query for the same information (with respect to a given model) by using an alternate template. We argue that this means we are measuring a *lower* bound for what language models know. We make this argument by analogy with traditional knowledge bases: they only have a *single* way of querying knowledge for a specific relation, namely by using the relation id of that relation, and this way is used to measure their accuracy. For example, if the relation ID is `works-For` and the user asks for `is-working-for`, the accuracy of the KG would be 0.

Single Token We only consider single token objects as our prediction targets. The reason we include this limitation is that multi-token decoding adds a number of additional tuneable parameters (beam size, candidate scoring weights, length normalization, n-gram repetition penalties, etc.) that obscure the knowledge we are trying to measure. Moreover, well-calibrated multi-token generation is still an active research area, particularly for bidirectional models (see *e.g.* Welleck et al. (2019)).

Object Slots We choose to *only* query object slots in triples, as opposed to subject or relation slots. By including reverse relations (e.g. `contains` and `contained-by`) we can also query subject slots. We do not query relation slots for two reasons. First, surface form realisations of relations will span several tokens, and as we discussed above, this poses a technical challenge that is not in the scope of this work. Second, even if we could easily predict multi-token phrases, relations can generally be expressed with many different wordings, making it unclear what the gold standard pattern for a relation should be, and how to measure accuracy in this context.

Intersection of Vocabularies The models that we considered are trained with different vocabularies. For instance, ELMo uses a list of ∼800K tokens while BERT considers only ∼30K tokens. The size of the vocabulary can influence the performance of a model for the LAMA probe. Specifically, the larger the vocabulary the harder it would be to rank the gold token at the top. For this reason we considered a common vocabulary of ∼21K case-sensitive tokens that are obtained from the intersection of the vocabularies for all considered models. To allow a fair comparison, we let every model rank only tokens in this joint vocabulary.

5 Results

We summarize the main results in Table 2, which shows the mean precision at one (P@1) for the different models across the set of corpora considered. In the remainder of this section, we discuss the results for each corpus in detail.

Google-RE We query the LMs using a standard cloze template for each relation. The base and large versions of BERT both outperform all other models by a substantial margin. Furthermore, they

Corpus	Relation	Statistics		Baselines		KB		LM					
		#Facts	#Rel	Freq	DrQA	RE_n	RE_o	Fs	Txl	Eb	E5B	Bb	Bl
Google-RE	`birth-place`	2937	1	4.6	-	3.5	13.8	4.4	2.7	5.5	7.5	14.9	**16.1**
	`birth-date`	1825	1	1.9	-	0.0	**1.9**	0.3	1.1	0.1	0.1	1.5	1.4
	`death-place`	765	1	6.8	-	0.1	7.2	3.0	0.9	0.3	1.3	13.1	**14.0**
	Total	5527	3	4.4	-	1.2	7.6	2.6	1.6	2.0	3.0	9.8	**10.5**
T-REx	1-1	937	2	1.78	-	0.6	10.0	17.0	36.5	10.1	13.1	68.0	**74.5**
	N-1	20006	23	23.85	-	5.4	**33.8**	6.1	18.0	3.6	6.5	32.4	34.2
	N-M	13096	16	21.95	-	7.7	**36.7**	12.0	16.5	5.7	7.4	24.7	24.3
	Total	34039	41	22.03	-	6.1	**33.8**	8.9	18.3	4.7	7.1	31.1	32.3
ConceptNet	Total	11458	16	4.8	-	-	-	3.6	5.7	6.1	6.2	15.6	**19.2**
SQuAD	Total	305	-	-	**37.5**	-	-	3.6	3.9	1.6	4.3	14.1	17.4

Table 2: Mean precision at one (P@1) for a frequency baseline (Freq), DrQA, a relation extraction with naïve entity linking (RE_n), oracle entity linking (RE_o), fairseq-fconv (Fs), Transformer-XL large (Txl), ELMo original (Eb), ELMo 5.5B (E5B), BERT-base (Bb) and BERT-large (Bl) across the set of evaluation corpora.

obtain a 2.2 and 2.9 respective average accuracy improvement over the oracle-based RE baseline. This is particularly surprising given that with the gold-aligned Google-RE source we know for certain that the oracle RE baseline has seen at least one sentence expressing each test fact. Moreover, the RE baseline was given substantial help through an entity linking oracle.

It is worth pointing out that while BERT-large does better, this does not mean it does so for the right reasons. Although the aligned Google-RE sentences are likely in its training set (as they are part of Wikipedia and BERT has been trained on Wikipedia), it might not "understand" them to produce these results. Instead, it could have learned associations of objects with subjects from co-occurrence patterns.

T-REx The knowledge source derived from Google-RE contains relatively few facts and only three relations. Hence, we perform experiments on the larger set of facts and relations in T-REx. We find that results are generally consistent with Google-RE. Again, the performance of BERT in retrieving factual knowledge are close to the performance obtained by automatically building a knowledge base with an off-the-shelf relation extraction system and oracle-based entity linking. Broken down by relation type, the performance of BERT is very high for 1-to-1 relations (*e.g.*, *capital of*) and low for N-to-M relations.

Note that a downstream model could learn to make use of knowledge in the output representations of a language model even if the correct answer is not ranked first but high enough (*i.e.* a hint about the correct answer can be extracted from the output representation). Figure 2 shows the mean P@k curves for the considered models. For BERT, the correct object is ranked among the top ten in around 60% of the cases and among the top 100 in 80% of the cases.

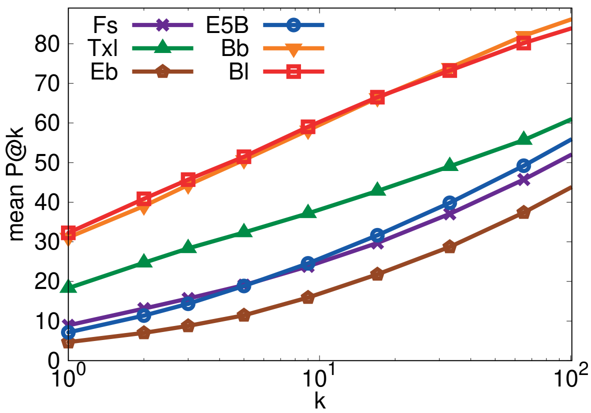

Figure 2: Mean P@k curve for T-REx varying k. Base-10 log scale for X axis.

To further investigate why BERT achieves such strong results, we compute the Pearson correlation coefficient between the *P@1* and a set of metrics that we report in Figure 3. We notice, for instance, that the number of times an object is mentioned in the training data positively correlates with performance while the same is not true for the subject of a relation. Furthermore, the log probability of a prediction is strongly positively correlated with P@1. Thus, when BERT has a high confidence in its prediction, it is often correct. Performance is also positively correlated with the cosine similarity between subject and object vectors, and

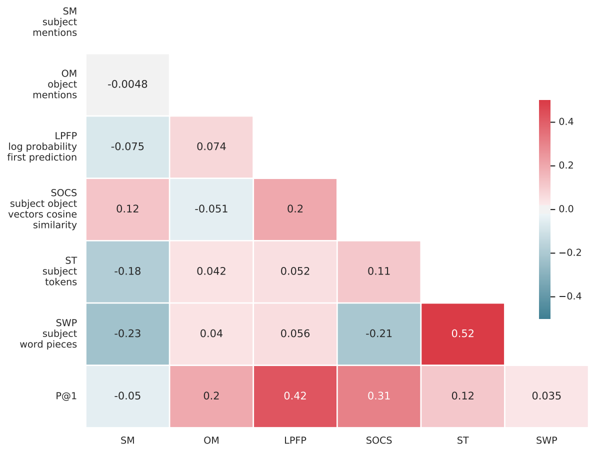

Figure 3: Pearson correlation coefficient for the P@1 of the BERT-large model on T-REx and a set of metrics: SM and OM refer to the number of times a subject and an object are mentioned in the BERT training corpus[4]respectively; LPFP is the log probability score associated with the first prediction; SOCS is the cosine similarity between subject and object vectors (we use spaCy[5]); ST and SWP are the number of tokens in the subject with a standard tokenization and the BERT WordPiece tokenization respectively.

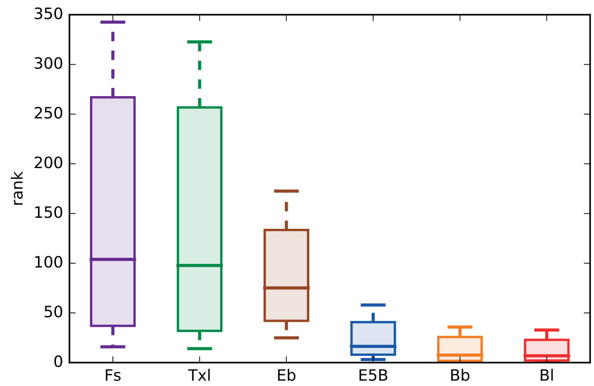

Figure 4: Average rank distribution for 10 different mentions of 100 random facts per relation in T-REx. ELMo 5.5B and both variants of BERT are least sensitive to the framing of the query but also are the most likely to have seen the query sentence during training.

slightly with the number of tokens in the subject.

Table 3 shows randomly picked examples for the generation of BERT-large for cloze template queries. We find that BERT-large generally predicts objects of the correct type, even when the predicted object itself is not correct.

To understand how the performance of a pretrained language model varies with different ways of querying for a particular fact, we analyze a maximum of 100 random facts per relation for which we randomly select 10 aligned sentences in Wikipedia from T-REx.[6] In each of the sentences, we mask the object of the fact, and ask the model to predict it. For several of our language models this also tests their ability to memorize and recall sentences from the training data since as the models have been trained on Wikipedia (see Table 1).

Figure 4 shows the average distribution of the rank for ten queries per fact. The two BERT models and ELMo 5.5B exhibit the lowest variability while ranking the correct object close to the top on average. Surprisingly, the performance of ELMo original is not far from BERT, even though this model did not see Wikipedia during training. Fairseq-fconv and Transformer-XL experience a higher variability in their predictions. Note that BERT and ELMo 5.5B have been trained on a larger portion of Wikipedia than fairseq-fconv and Transformer-XL and may have seen more sentences containing the test queries during training.

ConceptNet The results on the ConceptNet corpus are in line with those reported for retrieving factual knowledge in Google-RE and T-REx. The BERT-large model consistently achieves the best performance, and it is able to retrieve commonsense knowledge at a similar level to factual knowledge. The lower half of Table 3 shows generations by BERT-large for randomly sampled examples. Some of the concepts generated by the language models are surprisingly reasonable in addition to being syntactically correct.

SQuAD Next we evaluate our system on open-domain cloze-style question answering and compare against the supervised DrQA model. Table 2 shows a performance gap between BERT-large and the DrQA open-domain QA system on our cloze SQuAD task. Again, note that the pretrained language model is completely unsupervised, it is not fine-tuned, and it has no access to a dedicated information retrieval system. Moreover, when comparing DrQA and BERT-large in terms of P@10, we find that gap is remarkably small (57.1 for BERT-large and 63.5 for DrQA).

6 Discussion and Conclusion

We presented a systematic analysis of the factual and commonsense knowledge in publicly available pretrained language models *as is* and found

[5]The original training corpus is not available, we created our version using the same sources.

[5]https://spacy.io

[6]We exclude all facts with less than 10 alignments.